

Практикум D5

97 marks of the archive, six techniques, twelve tasks. Everything the archive asks about a normally distributed variable, from May 2021 to November 2025: an area, areas that add up without a calculator, a boundary found from an area, the mean or the standard deviation found from one area or two, and a condition inside.

О практикуме

Нормальная половина

statistics.continuous_random_variables: 103 балла из 199 (доля 0.518) — всё, где у величины кривая с μ и σ , а не плотность формулой. И 4 балла statistics.probability (0.017 — 4 из 237): сумма трёх площадей ноября 2023 Q10(a). 107 баллов по корпусу, 97 рабочих: десять — ноябрьский дубль 2023 года. 80 баллов Paper 2, 10 Paper 3, 7 Paper 1; доля калькулятора 90 из 97 честная.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-d5-normal

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

D5 — The normal distribution, forwards and backwards

97 marks of the archive, six techniques, twelve tasks. Everything the archive asks about a normally distributed variable, from May 2021 to November 2025: an area, areas that add up without a calculator, a boundary found from an area, the mean or the standard deviation found from one area or two, and a condition inside.

Almost all of it is on Paper 2. The calculator really is needed — and still does only the last step.

The one idea

A normal variable has no table. It has a **curve**, and

a probability is the area under the curve, above the values where the event happens.

$P(X < a)$ is the area to the left of a ; $P(a < X < b)$ the area between; $P(X > b)$ the area to the right. There is no area above a single value, so $P(X \leq a) = P(X < a)$: the markscheme does not care about $<$ against $\leq$, and neither do the checks.

And the idea that carries the rest

Every harder question is the same area **read backwards**. The calculator's `normalcdf` goes from a boundary to an area; `invNorm` goes from an area to a boundary. When the boundary is known and σ is not, the standardised boundary

$$z = \frac{x - \mu}{\sigma}$$

is the bridge: `invNorm` gives z , and $x = \mu + z\sigma$ is one equation for the unknown. Two unknowns need two areas.

How the checks work

They do not know the answers. Each check is handed **the model and the**

facts the question gives:

```
T = Normal(75, sigma**2, 'T')
verify_letters('10a', 3.41, sigma, [T],
[Eq(P(T > 82), 0.02)])
```

is *"your σ : does the area to the right of 82 come out as 0.02?"* The check finds σ itself, from the curve alone — no z-tables, no inverse normal inside — and compares. `Normal(mean, variance)` takes the **variance**, as IB writes $N(\mu, \sigma^2)$.

When you are wrong the check says **how**:

what you wrote	what the check says
the area on the other side	the opposite event
$P(X < 185)$ for $P(170 < X < 185)$	one boundary is lost
σ^2 typed where σ belongs	the second number of $N(\mu, \sigma^2)$ is the variance
the height of the curve	a probability is an area, not a height
$P(A \cap B)$ for $P(A B)$	not divided by the condition
$\sigma < 0$ from the equation	a standard deviation is positive
0.6827 when two figures are asked	the question asks for 2 significant figures

Order of work

level	what it means	tasks
	the area, and areas that add up without a calculator	1-3
	a boundary from an area; σ , then μ and σ	4-9
	a condition inside: given, a mixture, an event about another quantity	10-12

Every task is a real past-paper question, cited.

90 of these 97 marks are on a calculator paper, and the number is honest. Nobody integrates $e^{-x^2/2}$ by hand. But the method marks sit on the sketch and the sentence " $P(W < w) = 1 - 0.8 - 0.115$ ", not on the button.

```

import sys
sys.path.append('..')          # from
practicum/statistics to practicum/kit/
import sympy as sp             # the escape
hatch: anything not in kit is in sp
from kit import *              # checks +
Normal, Mix, Freq, P(), SD

language('en')                 # this
notebook is in English, and so are the checks

# Normal(mean, variance, name) is  $X \sim N(\text{mean}, \text{variance})$  – the variance, as IB writes it.
# Comparisons are events, P() finds their
area; & is "and", | is "or".
#      L = Normal(420, 30**2, 'L')
#      P(L < 400), P((L > 400) & (L < 460)),
P(L > 460, given=L > 420)
# Letters are allowed in the model: Normal(m,
s**2) – the checks find them.

print('ready; sympy', sp.__version__)
L = Normal(420, 30**2, 'L')
print('the model:                ', L)
print('P(L < 400):                ', sympify(P(L
< 400)))
print('P(400 < L < 460):          ',
sympify(P((L > 400) & (L < 460))))

```

Map of the six techniques

#	technique	you recognise it by	it reduces to
1	the area	μ and σ given; <i>find the probability, percentage, expected number</i>	<code>normalcdf</code> over the right interval
2	areas that add up	Paper 1; 95 % <i>within two standard deviations</i> ; areas given, one missing	symmetry, total = 1, the rule in the question
3	a boundary from an area	<i>find w such that, the top 20 %, quartile, k standard deviations</i>	<code>invNorm</code> from the area to the left
4	one parameter	σ (or μ) unknown, one area given	$z = \frac{x-\mu}{\sigma}$, one equation
5	two parameters	μ and σ both unknown, two areas given	two z equations, a linear system
6	a condition inside	<i>given that</i> , two kinds mixed, an	a ratio of areas; a sum of weighted

#	technique	you recognise it by	it reduces to
		event about another quantity	areas; an interval of Z

Technique 1 is the whole topic read forwards. Technique 2 is technique 1 with no calculator: the areas come from symmetry. Techniques 3–5 read it backwards, and each adds one unknown. Technique 6 puts a second question around the area.

Part 1. The area

Theory: an area over the right interval

A battery lasts L hours, with $L \sim N(420, 30^2)$.

Draw it first. A bell centred at 420, and the region the question names shaded. Then the calculator:

question	region	calculator
lasts less than 400 hours	left of 400	<code>normalcdf(-1E99, 400, 420, 30) = 0.252</code>
between 400 and 460 hours	between	<code>normalcdf(400, 460, 420, 30) = 0.656</code>
more than 460 hours	right of 460	<code>normalcdf(460, 1E99, 420, 30) = 0.0912</code>

The calculator wants the **standard deviation**, 30 — not the variance 900. The model is written $N(\mu, \sigma^2)$, and the second number in it is squared.

"More than" and "at least" are the same area. A single value has no area, so $P(L \geq 460) = P(L > 460)$. This is the

opposite of D3, where $X \geq 10$ and $X > 10$ differed by a whole bar.

Expected number. Of 250 batteries, the expected number lasting more than 460 hours is $250 \times 0.0912 \dots = 22.8$. It is a mean, so it need not be whole — but "*estimate the number*" is also answered with the whole number 23.

A percentage is the same area times 100 : 9.12 %.

Three significant figures, not three decimal places. 0.0912 has three significant figures; 0.091 has two, and on Paper 2 that is an A0.

Task 1 ● — **May 2025 TZ2 Paper 2**
Q10(a), (c), 4 marks

At Adam's Apple Orchard the weights of apples, W , in grams, are normally distributed with a mean 175 grams and standard deviation 8 grams.

(a) Find the probability that a randomly chosen apple weighs less than 170 grams.

All orchards classify an apple as premium when its weight is between 170 and 185 grams.

(c) Find the percentage of apples that are classified as premium at Adam's Apple Orchard.

Part (b) of this question is a boundary, and part (f) has two unknowns — they are in the archive.

```
qla = ...          # P(W < 170)
qlc = ...          # the percentage of premium
apples

W = Normal(175, 8**2, 'W')

verify_chance('la', qla, P(W < 170))
verify_chance('lc', qlc, P((W > 170) & (W <
185)), percent=True)
```

Theory: areas that add up without a calculator

On Paper 1 there is no `normalcdf`, and the question is built so that you need none. Three facts do the work.

Symmetry. The curve is symmetric about μ : half the area is on each side,

and the area beyond $\mu + d$ equals the area below $\mu - d$.

The total is one. If $P(X < a) = 0.3$ and $P(X > b) = 0.1$, then $P(a < X < b) = 1 - 0.3 - 0.1 = 0.6$ — no model needed at all.

The rule for whole standard deviations. For every normal curve,

within	area
$\mu \pm \sigma$	≈ 0.68
$\mu \pm 2\sigma$	≈ 0.95
$\mu \pm 3\sigma$	≈ 0.997

For $X \sim N(50, 4^2)$: 62 is three standard deviations above the mean, so by symmetry $P(X > 62) \approx \frac{1-0.997}{2} = 0.0015$. And an interval that is not symmetric is two halves: 46 is one standard deviation below, 58 two above, so $P(46 < X < 58) \approx \frac{0.68}{2} + \frac{0.95}{2} = 0.815$.

When the question gives the rule, use the question's rule. *"You may assume 95 % lie within two standard deviations"* means 0.95 exactly, even though the curve gives 0.9545. The

markscheme's answer is built on the question's number.

The same number of standard deviations, the same area. If $X \sim N(50, \sigma^2)$ and $Y \sim N(80, \sigma^2)$, then $P(Y > 86) = P(X > b)$ needs b to be as far above 50 as 86 is above 80: $b = 56$, whatever σ is. With different σ you compare $\frac{x-\mu}{\sigma}$ instead of $x - \mu$.

Task 2 ● — **May 2025 TZ3 Paper 1**
Q5, 6 marks

The random variables X and Y are normally distributed with $X \sim N(7, a^2)$ and $Y \sim N(19, a^2)$, where $a > 0$.

- (a)** Find b such that $P(X > b) = P(Y > 22)$.
- (b)** Write down the approximate value of $P(7 - a < X < 7 + a)$, correct to two significant figures.
- (c)** Given that $a = 3$, calculate the approximate value of $P(Y < 22)$, correct to two significant figures.

Paper 1. The checks for (a) and (b) try several values of a : your answer must not

depend on it.

```
q2a = ...          # b
q2b = ...          # P(7 - a < X < 7 + a), two
significant figures
q2c = ...          # P(Y < 22) when a = 3, two
significant figures

a, b = symbols('a b')
X = Normal(7, a**2, 'X')
Y = Normal(19, a**2, 'Y')

verify_letters('2a', q2a, b, [X, Y], [Eq(P(X
> b), P(Y > 22))], free=a)
verify_chance('2b', q2b, P((X > 7 - a) & (X <
7 + a)), sf=2, free=a)
verify_chance('2c', q2c, P(Normal(19, 3**2,
'Y') < 22), sf=2)
```

Task 3 — May 2024 TZ2 Paper 1 Q6, 5 marks

A farmer grows two types of apples — cooking apples and eating apples. The weights of the apples, in grams, can be modelled as normal distributions with the following parameters.

apple type	mean	standard deviation
eating	100 g	20 g
cooking	140 g	40 g

For each type of apple you can assume that 95% of the weights are within two standard deviations of the mean.

(a) Find the percentage of eating apples that have a weight greater than 140 g.

The farmer grows a large number of apples of which 80% are eating apples. Both types of apples are picked and randomly mixed together in a cleaning machine. After cleaning, the machine separates out those that have a weight greater than 140 g into a container.

(b) An apple is randomly selected from this container. Find the probability that it is an eating apple. Give your answer in the form $\frac{c}{d}$, where $c, d \in \mathbb{Z}^+$.

rule={2: 0.95} is the question's own sentence: the checks take areas from it, not from the curve. Part (b) is technique 6 on Paper 1 — try it now, it is fractions; Mix is explained in Part 3.

```
q3a = ...          # the percentage of eating
apples over 140 g
q3b = ...          # P(eating | over 140 g), as
Rational(c, d)

eating = Normal(100, 20**2, 'eating', rule=
{2: 0.95})
cooking = Normal(140, 40**2, 'cooking', rule=
{2: 0.95})
apple = Mix({eating: 0.8, cooking: 0.2},
'apple')

verify_chance('3a', q3a, P(eating > 140),
percent=True)
verify_chance('3b', q3b,
P(apple.came_from(eating), given=apple >
140))
```

Part 2. Backwards: the boundary, then the parameters

Theory: a boundary from an area

`invNorm(area,  $\mu$ ,  $\sigma$ )` returns the value with
that area **to its left**. Everything else is
translated into that first:

the question says	area to the left	for $L \sim N(420, 30^2)$
10 % last less than w	0.1	$w = 381.6$
10 % last more than w	$1 - 0.1 = 0.9$	$w = 458.4$
the lower quartile	0.25	$Q_1 = 399.8$
the upper quartile	0.75	$Q_3 = 440.2$

The quartiles are symmetric about the
mean, so the interquartile range is $Q_3 -$
 $Q_1 = 40.5$.

**When the area is not given directly,
find it first.** If 70 % of the batteries last

between w and 460 hours, the area to the left of w is $1 - 0.7 - P(L > 460) = 1 - 0.7 - 0.0912 = 0.2088$ — technique 2 — and only then `invNorm`.

" k standard deviations above the mean" is the boundary $\mu + k\sigma$. For every normal curve the area beyond it is the same, so k is read from $N(0, 1)$: the area above k is 0.05 when $k = 1.645$.

Four significant figures when asked.

"Correct to four significant figures" is a separate instruction, and the markscheme writes *must be 4 sf*. 181.73 is not an answer to it.

Task 4 — **May 2023 TZ2 Paper 2 Q3(a)-(c), 6 marks**

The weights, W grams, of bags of rice packaged in a factory can be modelled by a normal distribution with mean 204 grams and standard deviation 5 grams.

(a) A bag of rice is selected at random. Find the probability that it weighs more than 210 grams.

According to this model, 80 % of the bags of rice weigh between w grams and 210 grams.

(b) Find the probability that a randomly selected bag of rice weighs less than w grams.

(c) Find the value of w .

The checks for (b) and (c) find w themselves from the 80 % sentence, so a slip in (a) costs (a) only.

```
q4a = ...          # P(W > 210)
q4b = ...          # P(W < w)
q4c = ...          # w

w = symbols('w')
W = Normal(204, 5**2, 'W')
eighty = [Eq(P((W > w) & (W < 210)), 0.8)]

verify_chance('4a', q4a, P(W > 210))
verify_chance('4b', q4b, P(W < w),
given=eighty, var=w)
verify_letters('4c', q4c, w, [W], eighty)
```

Task 5 — **November 2025 TZ1** **Paper 2 Q5, 5 marks**

Consider the following grouped data set.

class	frequency
$15 < y \leq 20$	31
$20 < y \leq 25$	42
$25 < y \leq 30$	61
$30 < y \leq 35$	46
$35 < y \leq 40$	29

(a) Find an estimate for **(i)** the mean; **(ii)** the standard deviation.

The data can be approximated by a normal distribution.

(b) Use this information and your mean and standard deviation found in part (a) to find an approximate value for the interquartile range correct to two significant figures.

The quartile letters in the check are just names: the check finds them from the model it builds out of the table.

```

q5m = ...           # the mean
q5s = ...           # the standard deviation
q5b = ...           # the interquartile range,
two significant figures

data = Freq({Interval.Lopen(15, 20): 31,
Interval.Lopen(20, 25): 42,
                Interval.Lopen(25, 30): 61,
Interval.Lopen(30, 35): 46,
                Interval.Lopen(35, 40): 29},
'y')
Y = Normal(Expect(data), Var(data), 'Y')
Q1, Q3, R = symbols('Q1 Q3 R')
quartiles = [Eq(P(Y < Q1), 0.25), Eq(P(Y <
Q3), 0.75), Eq(R, Q3 - Q1)]

verify_moment('5a(i)', q5m, Expect(data))
verify_moment('5a(ii)', q5s, SD(data))
verify_letters('5b', q5b, R, [Y], quartiles,
sf=2)

```

Theory: one unknown parameter

When σ is unknown, `invNorm` cannot take it as an argument. Go through $Z \sim N(0,1)$ instead: the same area sits to the left of z on the standard curve and to the left of x on yours, and

$$z = \frac{x - \mu}{\sigma} \implies x = \mu + z\sigma$$

One area, one boundary. $X \sim N(90, \sigma^2)$ and $P(X > 100) = 0.05$. The area to the left of 100 is 0.95, so $z = 1.645$ (from `invNorm(0.95)`), and

$$\frac{100 - 90}{\sigma} = 1.6449 \implies \sigma = 6.08$$

A symmetric interval. 90% lie within 12 of the mean 60: the two tails share 10%, so 5% is above 72, the area to the left of 72 is 0.95, and $\frac{12}{\sigma} = 1.6449$ gives $\sigma = 7.30$. Halve the outside, never the inside.

An interquartile range. The quartiles are $\mu \pm 0.6745\sigma$ (the z with area 0.75 to its left is 0.6745), so an interquartile range of 16 means $2 \times 0.6745\sigma = 16$ and $\sigma = 11.9$.

The calculator can also just solve it.

Graph $y = \text{normalcdf}(-10^{99}, 100, 90, x)$ against $y = 0.95$ and intersect: the markscheme accepts that route with a sketch. It is the same equation.

A negative root never survives: a standard deviation is positive.

Task 6 ● — *May 2023 TZ1 Paper 2*
Q4, 5 marks

A company manufactures metal tubes for bicycle frames. The diameters of the tubes, D mm, are normally distributed with mean 32 and standard deviation s . The interquartile range of the diameters is 0.28.

Find the value of s .

```
q6 = ...          # s

s, Q1, Q3 = symbols('s Q1 Q3')
D = Normal(32, s**2, 'D')
spread = [Eq(P(D < Q1), 0.25), Eq(P(D < Q3),
0.75), Eq(Q3 - Q1, 0.28)]

verify_letters('6', q6, s, [D], spread)
```

Task 7 ● — *May 2025 TZ1 Paper 2*
Q4(c), 3 marks

The heights, H cm, of adults in a group can be modelled by a normal distribution with mean 163 cm and standard deviation s cm.

It is found that 88% of the group have a height between 153 cm and 173 cm.

(c) Find the value of s .

Parts (a) and (b) of this question are about regression lines — that is D7.

```
q7 = ...          # s

s = symbols('s')
H = Normal(163, s**2, 'H')

verify_letters('7', q7, s, [H], [Eq(P((H >
153) & (H < 173)), 0.88)])
```

Theory: two unknown parameters

Two areas, two boundaries, two equations. $X \sim N(\mu, \sigma^2)$ with $P(X < 30) = 0.2$ and $P(X > 45) = 0.1$:

- the area to the left of 30 is 0.2, so $z_1 = -0.8416$ and $\mu - 0.8416\sigma = 30$;
- the area to the left of 45 is $1 - 0.1 = 0.9$, so $z_2 = 1.2816$ and $\mu + 1.2816\sigma = 45$.

Subtract: $2.1232\sigma = 15$, so $\sigma = 7.06$ and $\mu = 35.9$.

The equations are in z , not in probabilities. $\mu + 0.2\sigma = 30$ is the most

common wrong system, and the markscheme's M1 says *"that involve z-values rather than probabilities"*.

The unknowns can be hidden in the story. *"The mean is 20 more and the standard deviation is half as big"* is a second model $N(\mu + 20, (\sigma/2)^2)$ with the **same** two letters — each area about either model is one more equation.

Areas can be given in relation to each other. *"Twice as many below a as above b "* and *"80 % between"* are two equations in the two tail areas: with x below and $2x$ above, $x + 0.8 + 2x = 1$. Solve for the areas first (technique 2), then turn each into a z .

Task 8 🟡 — **November 2023 TZ1**
Paper 2 Q10(a), (b), (d), 10 marks

A farmer is growing a field of wheat plants. The height, H cm, of each plant can be modelled by a normal distribution with mean m and standard deviation s .

It is known that $P(H < 94.6) = 0.288$ and $P(H > 98.1) = 0.434$.

(a) Find the probability that the height of a randomly selected plant is between 94.6 cm and 98.1 cm.

(b) Find the value of m and the value of s .

In another field, the farmer is growing the same variety of wheat, but is using a different fertilizer. The heights of these plants, F cm, are normally distributed with mean 98.6 and standard deviation d . The farmer finds the interquartile range to be 4.82 cm.

(d) Find the value of d .

Part (c) is binomial — it is in D3.

```

q8a = ...          # P(94.6 < H < 98.1)
q8b = [...]        # [m, s]
q8d = ...          # d

m, s, d, Q1, Q3 = symbols('m s d Q1 Q3')
H = Normal(m, s**2, 'H')
wheat = [Eq(P(H < 94.6), 0.288), Eq(P(H >
98.1), 0.434)]
F = Normal(98.6, d**2, 'F')
spread = [Eq(P(F < Q1), 0.25), Eq(P(F < Q3),
0.75), Eq(Q3 - Q1, 4.82)]

verify_chance('8a', q8a, P((H > 94.6) & (H <
98.1)), given=wheat, var=[m, s])
verify_letters('8b', q8b, [m, s], [H], wheat)
verify_letters('8d', q8d, d, [F], spread)

```

Task 9 — November 2025 TZ3 Paper 2 Q4, 7 marks

A garden centre sells seeds for two varieties of sunflowers: large and giant.

The heights of large sunflowers are normally distributed with mean m and standard deviation s . It is known that 25% of large sunflowers have a height of over 180 cm.

(a) Given that $m + As = 180$, where $A \in \mathbb{R}$, find the value of A .

The heights of giant sunflowers are also normally distributed. The giant sunflowers have a mean height that is 35 cm more than that of the large sunflowers and a standard deviation that is double that of the large sunflowers. It is known that 98% of giant sunflowers have a height greater than 180 cm.

(b) Determine m and s .

In (a) m and s are not known yet, and A does not depend on them: the check tries several s .

```
q9a = ...           # A
q9b = [...]         # [m, s]

m, s, A = symbols('m s A')
large = Normal(m, s**2, 'large')
giant = Normal(m + 35, (2*s)**2, 'giant')

verify_letters('9a', q9a, A, [large],
[Eq(P(large > 180), 0.25), Eq(m + A*s, 180)],
free=s)
verify_letters('9b', q9b, [m, s], [large,
giant],
[Eq(P(large > 180), 0.25),
Eq(P(giant > 180), 0.98)])
```

Part 3. A condition inside

Theory: given, mixed, and an event about another quantity

Given that. The condition is the new whole; the event is the part of it that also happens:

$$\begin{aligned}P(L > 460 \mid L > 420) &= \frac{P(L > 460 \text{ and } L > 420)}{P(L > 420)} \\&= \frac{0.0912}{0.5} \\&= 0.182\end{aligned}$$

Draw both regions. The numerator is the **overlap** — here the event lies inside the condition, so it is the event itself; when it does not (*"less than 470, given more than 440"*), the numerator is the area between 440 and 470.

Two kinds mixed. 70% of batteries are type A, $L \sim N(420, 30^2)$, and 30% type B, $N(380, 25^2)$. A battery picked at random lasts less than 400 hours with probability

$$0.7 \times 0.2525 + 0.3 \times 0.7881 = 0.413$$

— a tree with a normal area on each branch. Backwards (Bayes), the chance that a battery under 400 hours is type A is $\frac{0.7 \times 0.2525}{0.413} = 0.428$. In a check, `Mix({A: 0.7, B: 0.3}, 'battery')` is that tree, and `battery.came_from(A)` is the branch.

An event about another quantity. $Z \sim N(0, 1)$ and the equation $x^2 - 2Zx + 4 = 0$. Translate every statement into an interval of Z :

- real roots: $4Z^2 - 16 \geq 0$, so $Z \leq -2$ or $Z \geq 2$, with probability 0.0455;
- the larger root $Z + \sqrt{Z^2 - 4}$ is less than 5: this needs $Z < 5$ and $Z^2 - 4 < (5 - Z)^2$, that is $Z < 2.9$ — so, together with real roots, $Z \leq -2$ or $2 \leq Z < 2.9$.

Squaring an inequality is only safe when both sides are non-negative — check that, or you add an interval that does not belong.

Task 10 ● — May 2021 TZ2 Paper 2
Q10(a)-(c), 9 marks

The flight times, T minutes, between two cities can be modelled by a normal distribution with a mean of 75 minutes and a standard deviation of σ minutes.

(a) Given that 2% of the flight times are longer than 82 minutes, find the value of σ .

(b) Find the probability that a randomly selected flight will have a flight time of more than 80 minutes.

(c) Given that a flight between the two cities takes longer than 80 minutes, find the probability that it takes less than 82 minutes.

Parts (d) and (e) are binomial — D3. The checks for (b) and (c) find σ themselves.

```

q10a = ...      # sigma
q10b = ...      # P(T > 80)
q10c = ...      # P(T < 82 | T > 80)

sigma = symbols('sigma')
T = Normal(75, sigma**2, 'T')
two_percent = [Eq(P(T > 82), 0.02)]

verify_letters('10a', q10a, sigma, [T],
two_percent)
verify_chance('10b', q10b, P(T > 80),
given=two_percent, var=sigma)
verify_chance('10c', q10c, P(T < 82, given=T
> 80), given=two_percent, var=sigma)

```

Task 11 — **May 2022 TZ1 Paper 2** **Q11(a), (c), (d), 14 marks**

A bakery makes two types of muffins: chocolate muffins and banana muffins.

The weights, C grams, of the chocolate muffins are normally distributed with a mean of 62 g and standard deviation of 2.9 g.

(a) Find the probability that a randomly selected chocolate muffin weighs less than 61 g.

The weights, B grams, of the banana muffins are normally distributed with a

mean of 68 g and standard deviation of 3.4 g.

Each day 60% of the muffins made are chocolate. On a particular day, a muffin is randomly selected from all those made at the bakery.

(c) (i) Find the probability that the randomly selected muffin weighs less than 61 g. **(ii)** Given that a randomly selected muffin weighs less than 61 g, find the probability that it is chocolate.

The machine that makes the chocolate muffins is adjusted so that the mean weight of the chocolate muffins remains the same but their standard deviation changes to σ g. The machine that makes the banana muffins is not adjusted. The probability that the weight of a randomly selected muffin from these machines is less than 61 g is now 0.157.

(d) Find the value of σ .

Part (b) is binomial — D3.

```

q11a = ...          # P(C < 61)
q11ci = ...         # P(muffin < 61)
q11cii = ...        # P(chocolate | muffin < 61)
q11d = ...          # sigma

C = Normal(62, 2.9**2, 'C')
B = Normal(68, 3.4**2, 'B')
muffin = Mix({C: 0.6, B: 0.4}, 'muffin')

sigma = symbols('sigma')
adjusted = Normal(62, sigma**2, 'C')
now = Mix({adjusted: 0.6, B: 0.4}, 'muffin')

verify_chance('11a', q11a, P(C < 61))
verify_chance('11c(i)', q11ci, P(muffin < 61))
verify_chance('11c(ii)', q11cii,
P(muffin.came_from(C), given=muffin < 61))
verify_letters('11d', q11d, sigma,
[adjusted], [Eq(P(now < 61), 0.157)])

```

Task 12 — May 2024 TZ2 Paper 3 Q2(g)-(h), 10 marks

In parts (g) and (h), consider a randomly generated quadratic function, $f(x) = x^2 + 2Zx + 1$, where the continuous random variable $Z \sim N(0, 1)$.

(g) Find the probability that the graph of f has two x -intercepts.

The continuous random variables, X_1 and X_2 , represent the x -intercepts of the

graph of f where $X_1 = -Z - \sqrt{Z^2 - 1}$ and $X_2 = -Z + \sqrt{Z^2 - 1}$.

(h) Given that the graph of f has two x -intercepts, X_1 and X_2 , find the probability that both X_1 and X_2 are greater than 0.5.

The check does not know the intervals of Z . It walks along the axis, works out X_1 and X_2 at each Z , and finds where the event starts and stops. Two distinct intercepts is `x1 < x2`: when $Z^2 < 1$ there are none, and at $Z^2 = 1$ they coincide.

```
q12g = ...      # P(two x-intercepts)
q12h = ...      # P(both > 0.5 | two x-
intercepts)

Z = Normal(0, 1, 'Z')
X1 = Z.map(lambda z: -z - sqrt(z**2 - 1),
'X1')
X2 = Z.map(lambda z: -z + sqrt(z**2 - 1),
'X2')
two = X1 < X2

verify_chance('12g', q12g, P(two))
verify_chance('12h', q12h, P((X1 > 0.5) & (X2
> 0.5), given=two))
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
area	the model is known: find an area
symmetry	no calculator: symmetry, total one, the whole- σ rule
boundary	the model is known: find the boundary from an area
sigma	one parameter unknown, one area given
both	the mean and the standard deviation both unknown
condition	given that, two kinds mixed, or an event about another quantity

1. Rods have lengths $N(150, 0.4^2)$ cm. Find the probability that a rod is longer than 150.5 cm.
2. Marks in a test are $N(62, 9^2)$; the top 5% get a distinction. Find the lowest mark for a distinction.
3. Bottles are filled with a mean of 500 ml, and 3% contain less than 495 ml.

Find the standard deviation.

4. $X \sim N(20, 3^2)$ and $P(X < 17) \approx 0.16$.
Without a calculator, write down $P(17 < X < 23)$.
5. 10% of parcels weigh less than 2 kg and 25% weigh more than 5 kg. Find the mean and the standard deviation.
6. Race times are $N(48, 5^2)$ minutes. Given that a runner finishes in under 50 minutes, find the probability that they finish in under 45.
7. Egg weights are $N(60, 4^2)$ g. Estimate how many of 300 eggs weigh between 55 and 65 g.
8. A normal distribution has mean 12 and interquartile range 3. Find σ .
9. 30% of a shop's bananas come from a farm with weights $N(120, 10^2)$ g, the rest from a farm with $N(130, 12^2)$ g. Find the probability that a banana weighs more than 125 g.
0. Find the upper quartile of $N(80, 6^2)$.
1. $X \sim N(\mu, \sigma^2)$ and $Y \sim N(\mu + 10, \sigma^2)$.
Find c such that $P(X < c) = P(Y < 25)$,

in terms of μ — without a calculator.

2. $P(X < 40) = 0.3$ and $P(X < 55) = 0.8$ for a normal X . Find its mean.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: '4a91ee5f0106', 2:
'931534c0c145', 3: '38de90475bb3', 4:
'3781049acb7f', 5: 'ff7772053abf', 6:
'596d40a4ed24', 7: '4a91ee5f0106', 8:
'38de90475bb3', 9: '596d40a4ed24', 10:
'931534c0c145', 11: '3781049acb7f', 12:
'ff7772053abf'})
```

On the clock — May 2024 TZ1 Paper 2 Q3, 4 marks

Four marks, five minutes. Calculator allowed, no hints.

The random variable X is normally distributed with mean 10 and standard deviation 2.

- (a)** Find the probability that X is more than 1.5 standard deviations above the

mean.

The probability that X is more than k standard deviations above the mean is 0.1, where $k \in \mathbb{R}$.

(b) Find the value of k .

Attempt log

date	time	result

```
qt_a = ...          # P(X more than 1.5 standard
deviations above the mean)
qt_b = ...          # k

k = symbols('k')
X = Normal(10, 2**2, 'X')

verify_chance('timer (a)', qt_a, P(X > 10 +
1.5*2))
verify_letters('timer (b)', qt_b, k, [X],
[Eq(P(X > 10 + k*2), 0.1)])
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a) The region left of 170: $P(W < 170) = \text{normalcdf}(-10^{99}, 170, 175, 8) = 0.265985 \dots =$
0.266

1 (c) $P(170 < W < 185) = 0.628364 \dots$, so
62.8 %.

The markscheme accepts 26.6 % in (a) as well; a probability is still the better form there.

2 (a) 22 is 3 above the mean of Y , and both curves have the same standard deviation a : the same area lies above $7 + 3 \cdot \frac{b-7}{a}$. The markscheme's line is $\frac{b-7}{a} = \frac{22-19}{a}$.

2 (b) $\mu \pm \sigma$ holds about 68 %: 0.68.

2 (c) With $a = 3$, $22 = 19 + 3$ is one standard deviation above the mean. Above it lies $\frac{1-0.68}{2} = 0.16$, so

$$P(Y < 22) \approx 1 - 0.16 = \boxed{0.84}$$

3 (a) $140 = 100 + 2 \times 20$. Outside $\mu \pm 2\sigma$ is 5%, split equally: $\boxed{2.5\%}$ above.

3 (b) For cooking apples 140 **is** the mean, so half of them are over 140. The container holds

$$0.8 \times 0.025 + 0.2 \times 0.5 = 0.02 + 0.1 = 0.12$$

of all apples, and the eating ones among them are 0.02:

$$P(\text{eating} | > 140) = \frac{0.02}{0.12} = \boxed{\frac{1}{6}}$$

4 (a) $P(W > 210) = 0.115069\dots = \boxed{0.115}$

4 (b) Below w , between, above 210 make up the whole:

$$\begin{aligned}
 P(W < w) &= 1 - 0.8 - 0.115069 \dots \\
 &= 0.0849302 \dots \\
 &= \boxed{0.0849}
 \end{aligned}$$

4 (c) $w = \text{invNorm}(0.0849302 \dots, 204, 5) = 197.136 \dots = \boxed{197}$

5 (a) Mid-interval values 17.5, 22.5, 27.5, 32.5, 37.5 with the frequencies, $\sum f = 209$: mean $\boxed{27.5}$, standard deviation $6.26374 \dots = \boxed{6.26}$ (the calculator's σ_x , which divides by n ; $s_x = 6.28$ divides by $n - 1$).

5 (b) With $Y \sim N(27.5, 6.26374 \dots^2)$: $Q_1 = \text{invNorm}(0.25) = 23.2751 \dots$ and $Q_3 = \text{invNorm}(0.75) = 31.7248 \dots$, so

$$\text{IQR} = 8.44965 \dots = \boxed{8.4}$$

6 Half the interquartile range each side of the mean: $Q_3 = 32 + 0.14 = 32.14$, with area 0.75 to its left. So $z = 0.674489 \dots$ and

$$\begin{aligned}
\frac{32.14 - 32}{s} &= 0.674489 \dots \\
&\implies s \\
&= 0.207564 \dots \\
&= \boxed{0.208} \text{ mm}
\end{aligned}$$

7 The interval is 163 ± 10 , symmetric: 6% below 153 and 6% above 173. The area to the left of 173 is 0.94, $z = 1.55477 \dots$, and

$$\begin{aligned}
\frac{10}{s} &= 1.55477 \dots \\
&\implies s \\
&= 6.43181 \dots \\
&= \boxed{6.43}
\end{aligned}$$

8 (a) $P(94.6 < H < 98.1) = 1 - 0.288 - 0.434 = \boxed{0.278}$

8 (b) Areas to the left: 0.288 below 94.6, and $1 - 0.434 = 0.566$ below 98.1. So $z_1 = -0.559236 \dots$, $z_2 = 0.166199 \dots$:

m

$$\begin{aligned} & - 0.559236 \dots s = 94.6, & m \\ & + 0.166199 \dots s \\ & = 98.1 \end{aligned}$$

Subtracting, $0.725436 \dots s = 3.5: \quad s = 4.82468 \dots, \quad m = 97.2981 \dots$

$$\boxed{m = 97.3, s = 4.82}$$

8 (d) $Q_3 = 98.6 + 2.41 = 101.01$ has area 0.75 to its left: $\frac{2.41}{d} = 0.674489 \dots$, so $d = 3.57307 \dots = \boxed{3.57}$.

9 (a) 25 % above 180 is 75 % below: $z = 0.674489 \dots$, and $180 = m + 0.674489 \dots s$ — which is $m + As = 180$ with $\boxed{A = 0.674}$.

9 (b) Giant: $N(m + 35, (2s)^2)$. 98 % above 180 is 2 % below, so $z = -2.05374 \dots$:

$$\begin{aligned} \frac{180 - (m + 35)}{2s} &= -2.05374 \dots \\ &\implies m - 4.10749 \dots s \\ &= 145 \end{aligned}$$

With $m + 0.674489 \dots s = 180$: $4.78198 \dots s = 35$, so $s = 7.31913 \dots$ and $m = 175.063 \dots$

$$\boxed{m = 175, s = 7.32}$$

10 (a) 2% above 82 is 98% below: $z = 2.05374\dots$, and

$$\begin{aligned}\frac{82 - 75}{\sigma} &= 2.05374\dots \\ &\implies \sigma \\ &= 3.40840\dots \\ &= \boxed{3.41}\end{aligned}$$

10 (b) $P(T > 80) = 0.0711929\dots = \boxed{0.0712}$

10 (c) The overlap of "*less than 82*" and "*more than 80*" is between 80 and 82:

$$\begin{aligned}P(T < 82 \mid T > 80) &= \frac{P(80 < T < 82)}{P(T > 80)} \\ &= \frac{0.0511929\dots}{0.0711929\dots} \\ &= 0.719073\dots \\ &= \boxed{0.719}\end{aligned}$$

11 (a) $P(C < 61) = 0.365111\dots = \boxed{0.365}$

11 (c)(i) $P(B < 61) = 0.0197555\dots$, and on the tree

$$\begin{aligned}0.6 \times 0.365111\dots + 0.4 \times 0.0197555\dots &= 0.226969\dots \\ &= \boxed{0.227}\end{aligned}$$

$$\mathbf{11 (c)(ii)} \quad \frac{0.6 \times 0.365111 \dots}{0.226969 \dots} = 0.965183 \dots = \boxed{0.965}$$

The markscheme wants the recognition *in context* — “ $P(\text{chocolate} \mid \text{weight} < 61)$ ” — not just $P(A \mid B)$.

11 (d) The tree again, with the new chocolate area unknown:

$$\begin{aligned} 0.6 P(C < 61) + 0.4 \times 0.0197555 \dots &= 0.157 \\ &\implies P(C < 61) \\ &= 0.248496 \dots \end{aligned}$$

Then $z = -0.679229 \dots$ and $\frac{61-62}{\sigma} = -0.679229 \dots$: $\sigma = 1.47225 \dots = \boxed{1.47}$ g.

12 (g) Two intercepts when the discriminant is positive: $4Z^2 - 4 > 0$, so $Z < -1$ or $Z > 1$:

$$\begin{aligned} P(|Z| > 1) &= 1 - 0.682689 \dots \\ &= 0.317310 \dots \\ &= \boxed{0.317} \end{aligned}$$

12 (h) $X_2 > X_1$, so both exceed 0.5 when $X_1 > 0.5$: $-Z - \sqrt{Z^2 - 1} > 0.5$ means $\sqrt{Z^2 - 1} < -Z - 0.5$. The right side must

be positive, so $Z < -0.5$ — with two intercepts, $Z < -1$ — and squaring, $Z^2 - 1 < Z^2 + Z + 0.25$, so $Z > -1.25$. The event is $-1.25 < Z < -1$:

$$\begin{aligned}\frac{P(-1.25 < Z < -1)}{P(|Z| > 1)} &= \frac{0.0530054\dots}{0.317310\dots} \\ &= 0.167046\dots \\ &= \boxed{0.167}\end{aligned}$$

The markscheme gives separate marks for -1.25 and for the inequality; $-1.25 < Z$ alone loses the last of them.

Timer

(a) $10 + 1.5 \times 2 = 13$: $P(X > 13) = 0.0668072\dots = \boxed{0.0668}$

(b) The area above k standard deviations is the same on every normal curve: $P(Z > k) = 0.1$, so $k = \text{invNorm}(0.9) = 1.28155\dots = \boxed{1.28}$.